miRPanel

An integrated miRNA target identification and interpretation tool

Technical documentation for supervisory review

This document records what the tool does, how it is built, which release of every external resource it uses, what went wrong at each stage, how each problem was diagnosed and solved, and which decisions were taken and on what evidence. Every figure quoted here was measured on the system and is stored in the project database or in the manifest files under manifests/, so any number can be re-queried rather than taken on trust.

Working name: miRPanel (placeholder, set in src/config.py). **Repository:** github.com/aungshuman02-blip/mirna-tool. **Platform:** Debian Linux. **Scope:** human (Homo sapiens) only.

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1. Project overview and current state

miRPanel answers one question end to end: given a human microRNA, which messenger RNAs is it likely to target, what is the evidence for each, and what biology do those targets add up to.

The user supplies a miRNA by name or accession, optionally restricts the search to particular genes, and selects which analysis methods to run. The system scans transcripts, collects the results of every selected method, and presents them as one table in which each method keeps its own column, its own units and its own direction. From that table the user can run gene-set enrichment (GO and KEGG), view the targets as a network, and export everything as a tab-separated file for scripting, an Excel workbook for reading, or a printable HTML report.

Design principle that governs everything else

Four tools answer four different questions, and no single number can carry four answers. The seed analyser asks whether a canonical Watson–Crick match exists. RNAhybrid asks whether a duplex could physically form. TargetScan asks how much repression is expected given the sequence context, and separately whether evolution has preserved the site. miRTarBase asks whether anyone has tested the pair experimentally.

Their scores are measured in different units on different scales: a context++ score of −0.503 and a folding energy of −18.5 kcal/mol are not comparable quantities, and averaging them would invent a number that means nothing. The tool therefore never produces a combined score and never ranks genes across tools. Section 14 gives the concrete case where combining would have destroyed the most important result in the data set.

Current state

The following are complete and in use:

- Data layer: four external resources loaded, reconciled and version-stamped
- Four analysis tools: seed analyser, RNAhybrid, TargetScan lookup, miRTarBase evidence
- Job orchestration with a background worker and three queues
- HTTP API (FastAPI) with a 21-assertion test suite
- Web interface served from the same process, no build step
- Gene-set enrichment through R and clusterProfiler, stored and reproducible
- Target network view, interactive on the page and as a static figure in the report
- Export in three formats, all carrying identical provenance

Not yet built: the benchmark (scripts written, not run), the machine-learning model (awaiting supervisory decision on features and training data), differential expression analysis, and deployment. Section 19 covers each.

2. Scope: what is included, what was cut, and why

The project originally planned three species and five prediction methods. Both were narrowed on 9 September 2026 as a deliberate decision, not as a response to time pressure.

Item	Decision	Reason
Mouse and rat	Removed	MANE Select is human-only; mouse would need Ensembl Canonical instead, and no TargetScan release exists for rat. Two different transcript backbones in one tool would make positions incomparable.
miRanda	Removed	Not packaged in Debian, and it answers a question RNAhybrid already answers (alignment plus energy). A second opinion of the same kind was judged not worth a hand-built binary.
3′UTR-only scanning	Extended to all regions	Validated sites occur in the CDS. BCL2's strongest canonical miR-21 site is a 7mer-m8 in coding sequence, which a 3′UTR-only tool cannot see at all.
Precomputed seed table	Rejected	Measured: 72 Mnt of transcript sequence loads from SQLite in 0.06 s and a full transcriptome seed scan takes 0.17 s. Storing a miRNA × transcript matrix would be large, stale and unnecessary.

Governing storage principle: store what cannot be recomputed, compute what can. Sequences, published evidence and precomputed third-party predictions are stored because they come from outside. Seed matches are computed live because they are fast and deterministic. RNAhybrid results are cached after computation because they are slow.

3. Technology stack and exact versions

Every version below is recorded in the database and printed on every export, so a result can always be traced to the software that produced it.

Component	Version	Role
Python	3.13.5	Backend, orchestration, all four tool wrappers
FastAPI	0.141.1	HTTP API
Starlette	1.6.0	ASGI layer beneath FastAPI
Pydantic	2.13.5	Request validation
Uvicorn	0.52.4	ASGI server
httpx	0.28.1	In-process API testing

Component	Version	Role
SQLite	3.46.1	Single-file database, WAL mode
openpyxl	current	Excel export (added 13 Sep 2026)
R	4.5.0	Enrichment analysis
Bioconductor	3.22	R package repository
clusterProfiler	4.18.4	GO and KEGG over-representation testing
org.Hs.eg.db	3.22.0	Human gene annotation
miRBaseConverter	1.34.0	Historical miRNA name export
RNAhybrid	2.1.2-8 (Debian)	RNA–RNA duplex folding
vis-network	10.1.2	Network rendering in the browser (vendored)

Phases 1 to 5 — the seed analyser, the RNAhybrid wrapper, the data layer and the orchestrator — use the Python standard library alone. No third-party Python package is required to reproduce the core science.

Why RNAhybrid comes from the Debian archive

RNAhybrid is a 2006-era C program. Building it by hand produces a binary whose provenance cannot be stated in a methods section. The Debian package rnahybrid 2.1.2-8 installs /usr/bin/RNAhybrid, RNAcalibrate and RNAeffective, is version-pinned by the distribution, and is reproducible by anyone on the same release. This also removed the need for the Docker container originally planned for it.

Why the browser library is vendored, not loaded from a CDN

vis-network is stored in web/vendor/ and served by the application's own server rather than fetched from a content delivery network. Three reasons: the tool must work on a machine with no internet access; a CDN outage must not break a page that is otherwise entirely self-hosted; and a pinned local copy is the only version claim that stays true. Its MIT and Apache-2.0 licences are stored beside it.

4. Data sources: loading, reconciliation and the problems each one raised

Four external resources are loaded. Each was version-checked, reconciled against the others, and every input row accounted for. The raw downloads are kept in data/raw/ so the database can always be rebuilt from scratch.

4.1 MANE Select v1.5 — the transcript backbone

MANE (Matched Annotation from NCBI and EMBL-EBI) is a set of transcripts on which RefSeq and Ensembl agree exactly, one per gene. It is the coordinate system everything else is anchored to.

Loaded	Count
Genes	19,363
Transcripts	19,437 (19,363 MANE Select + 74 MANE Plus Clinical)
Regions (5′UTR, CDS, 3′UTR)	57,460
Gene aliases	76,673

Decision: scans use the 19,363 MANE Select transcripts only. Each of the 74 MANE Plus Clinical transcripts belongs to a gene that already has a Select transcript, so including them would list 74 genes twice with different coordinates. Verified: hsa-miR-21-5p gives 4,365 sites of 7mer or better across MANE Select, and the 74 extra transcripts contribute exactly 28 more, reproducing the 4,393 measured before the orchestrator existed.

Why one transcript per gene matters: a binding site position is only meaningful against the sequence it was measured on. Anchoring everything to a single agreed transcript per gene is what makes the seed analyser's position and RNAhybrid's position directly comparable.

4.2 miRBase Release 23 — the miRNA namespace

miRBase Release 23 (August 2026) provides 69,020 mature miRNAs. The tool keys miRNAs by accession (MIMAT…), never by name.

Problem: seeds move between releases

243 of 2,594 human miRNAs had their seed sequence change between miRBase 22 and 23. The cause is not a different miRNA but 5′ end re-annotation: the same window slid one nucleotide along the sequence. Because the seed is defined as positions 2–8, trimming or extending the 5′ end renumbers every position.

```
hsa-miR-146b-3p (MIMAT0004766)
```

```
TargetScan (v22 era)   GCCCTGT
miRBase R23            CCCTGTG
```

Solution: miRNAs are keyed by accession, which is stable across releases, and TargetScan is joined on seed family rather than on miRNA identity. Where a seed has changed, the row is flagged seed_matches_current = 0 and the interface marks it rather than hiding the discrepancy.

Problem: old names cannot be resolved from current data

miRTarBase identifies miRNAs by name, and much of its CLIP-seq data comes from 2010–2013 papers using pre-suffix names such as hsa-miR-122 rather than hsa-miR-122-5p. miRBase serves only releases 20–22 directly, and 299 of these bare names have both arms annotated in Release 23, so the arm cannot be inferred.

Solution: historical name-to-accession pairs were exported from miRBaseConverter 1.34.0, covering 22 miRBase releases from v6 (2005) to v22 (2018) — 3,486 unique human pairs, 774 of them not obtainable any other way. The effect on the miRTarBase load is quantified in section 4.4.

4.3 TargetScan v8.0 — precomputed predictions

TargetScan 8.0 (September 2021) contributes 7,501,712 human predictions across 2,064 seed families. It is used as a local lookup and never queried live.

Problem: which miRBase release is TargetScan built on?

This had to be established rather than assumed, because a mismatch would silently join the wrong predictions to the wrong miRNA.

Compared against	Seeds identical	Seeds different	Accession absent
miRBase R22 (2,656 human)	2,550	4	52
miRBase R23 (2,789 human)	2,264	238	104

Agreement with R22 is 2,550/2,554 = 99.84% where comparable. TargetScan v8.0 is a miRBase v22-era resource. Joining it to R23 by accession resolves 2,502 of 2,606 (96.0%); 104 accessions no longer exist in R23, of which 31 are recoverable through their seed family and 73 (2.8%) are not.

Problem: gene identifiers do not match

TargetScan's Gene_info file lists 28,352 human transcripts across 19,474 genes using versioned Ensembl identifiers. Three candidate join keys were measured before one was chosen:

Join key	Distinct TargetScan entries finding a MANE match
Ensembl transcript	12,875
Ensembl gene	18,187
Gene symbol	17,212

Solution: a resolver cascade — Ensembl gene, then gene symbol, then gene alias — reaching 18,593 of 19,474 (95.5%). Version suffixes are stripped before joining. Where 110 Entrez ids received more than one Ensembl gene id, the Ensembl gene whose representative transcript is the MANE transcript was kept, with ties broken by sorted Ensembl id so the load is reproducible.

Problem: TargetScan scores a different transcript

TargetScan predicts on its own representative transcript per gene, chosen by 3P-seq support. For the resolved genes, that representative is the MANE transcript for 11,372 (61.2%) and a different transcript for 7,189 (38.7%).

Solution: TargetScan results are joined and displayed at **gene level, never transcript level**, and no TargetScan site position is ever shown. For the 38.7% the underlying 3′UTR sequence differs from the one the other tools scan, so a position would not be comparable. TargetScan's hg19 genome-coordinate files are never used. Section 14 shows this 38.7% landing on a validated target and explains a result that would otherwise look like a false negative.

Load accounting

Outcome	Rows
Loaded	7,501,712
Dropped — gene not in MANE	226,824
Dropped — duplicate Ensembl gene for one Entrez	36,529
Input total (Summary_Counts.all_predictions, human)	7,765,065

The three outcomes sum exactly to the input. Every row is accounted for; nothing is silently discarded.

4.4 miRTarBase 10.0 — experimental evidence

miRTarBase supplies the experimental record: 4,004,479 human interaction rows. It is classified as evidence, not prediction, and is never presented as a predictor.

A version trap worth noting

The download page is headed "Release 11.0", but every link serves /10.0/ files and /11.0/ returns 404. The database therefore records 10.0, which is what is actually loaded. A methods section quoting 11.0 from the page heading would be wrong.

Evidence tiers, and why one of them is not validation

Support type	Rows
Functional MTI (Weak)	3,981,769
Functional MTI	18,044
Non-Functional MTI	4,236
Non-Functional MTI (Weak)	430

Non-Functional MTI records a published finding that the miRNA does *not* regulate the gene. It is stored and counted but never displayed as validation. The loader refuses to run if any support type outside these four appears, so a future release that adds a category cannot be loaded silently.

Problem: 22.5% of evidence rows could not be resolved

Progressive reconciliation using the historical alias export from section 4.2:

Stage	Unresolved evidence rows
miRBase 22 aliases only	899,902 (22.5%)
Plus releases 21 and 20	899,902 (no improvement)
Plus miRBaseConverter v6–v22	184,409 (4.6%)

Of the 184,409 residual rows, 149,049 (83 miRNAs) are miRNAs *deleted* in miRBase Release 23. These are correctly unresolvable and are reported as retired rather than as absent — a meaningful distinction, because "retired" is a fact about the namespace and "absent" would be a fact about the data. A further 18 human aliases map to more than one accession across releases; the resolver reports these as ambiguous and refuses to guess.

Loaded

- mirtarbase_evidence — 4,004,479 rows
- mirtarbase_sites — 669,164 rows (target sites are sequences, not coordinates)
- mirtarbase — 1,612,045 summary pairs: 8,575 with strong functional support, 2,120 with contradicting strong evidence, 278,452 carrying a curated site

5. Database design

A single SQLite file, data/processed/mirna.db, approximately 1.6 GB. Opened in WAL (write-ahead logging) mode with busy_timeout = 15000 milliseconds, so readers never block the background worker and a brief lock contention waits rather than failing.

The schema is applied on every connection through src/db.py, which means a new table appears automatically without a migration step, and the worker can be started before the API has ever run.

Reference tables

Table	Holds	Key design point
data_sources	One row per downloaded file: version, URL, date, size, SHA-256	The single source of truth for versions. Every export and the page footer read from here, so a file and the page that produced it can never claim different releases.
genes	19,363 genes: Entrez, symbol, Ensembl, HGNC	Entrez id is the primary key throughout the system
transcripts	19,437 transcripts including full sequence	Sequence stored in the database, not on disk, so a scan is one query
regions	57,460 rows: 5′UTR, CDS, 3′UTR bounds per transcript	Lets any position be labelled with its region without re-deriving CDS bounds
gene_aliases	76,673 alternative gene names	So a user typing an obsolete symbol still finds the gene
mirnas	69,020 mature miRNAs with sequence and seed	Keyed by MIMAT accession
mirna_aliases	91,657 aliases spanning miRBase v6 to v23	The historical bridge described in section 4.2

Third-party prediction and evidence tables

Table	Rows	Note
targetscan_mirnas	2,580	26 TargetScan 5′ isoform entries skipped: no miRBase accession, so unbridgeable
targetscan_genes	18,357	11,320 (61.7%) sit on the MANE transcript
targetscan	7,501,712	2,064 distinct seeds; 107,790 flagged in TargetScan's default set
mirtarbase_evidence	4,004,479	Row-level published evidence
mirtarbase_sites	669,164	Curated target site sequences
mirtarbase	1,612,045	Summary pair table used at query time

Analysis tables

Three analysis subsystems — jobs, enrichment runs and reports — all follow the same shape: one row describing the analysis, many rows of output. The shape is deliberate and repeated so a new analysis type has an obvious template.

Table	Purpose	Notable columns
jobs	One miRNA against one target set	params (JSON: miRNA, tools, genes), status, batch_id for grouping several miRNAs
tool_runs	One row per tool per job	status, n_results, seconds, provenance, message — records a failed tool rather than losing it
results	One row per binding site or evidence record	tool, site_type, pos_start, pos_end, score, score_type, low_confidence, extra (JSON)
enrichment_runs	One row per enrichment analysis	gene_ids actually tested, background source and version, correction method, cutoffs, R and package versions, kegg_fetched_at
enrichment_terms	One row per enriched term	k, n, K, N — the four hypergeometric integers — plus p, adjusted p, q, and member genes
reports	One row per generated HTML report	spec (sort and limits), the rendered document, job_finished_at for staleness

Three principles the schema enforces

1. Every position is 1-based on the MANE transcript

A tool working on a sub-sequence converts inside its own wrapper before the record is stored. This is checked rather than trusted: all 45,034 RNAhybrid records from a transcriptome-wide run were verified by rebuilding the target strand from the alignment rows and comparing it against transcript[pos_start−1 : pos_end]. Zero mismatches.

2. Nothing is deleted when an analysis is re-run

An enrichment run stores the gene ids it actually tested, so it remains valid for those genes forever. job_finished_at records what the job looked like at the time, and the interface derives staleness by comparison rather than by destruction. A historical scientific result stays readable. The same rule now applies to generated reports.

3. Every p-value is recomputable without R

Each enriched term stores k (your genes in the term), n (your genes annotated in that ontology), K (background genes in the term) and N (background genes annotated in that ontology). Anyone can recompute the hypergeometric p-value in any language from the row alone. Section 11 explains why these four numbers are stored per term rather than once per run.

6. The tool contract: how a prediction method plugs in

Adding a fifth prediction method requires no change to the orchestrator, the API or the frontend. Every tool is a Python module in src/tools/ that defines three things.

```
META                  name, version, kind, score_type, better,
                      gives_position, needs_sequence, slow

covers(con, entrez)   can this tool say anything about this gene at all

run(con, mirna, targets, opts, progress)
                      -> {records, provenance, warnings}
```

META is validated at import time by check_meta(), so a tool with an incomplete or contradictory description cannot be registered at all.

Field	Meaning	Why it is mandatory
kind	prediction or evidence	miRTarBase is evidence and must never be presented as a predictor
score_type	The units, in words	Printed in the interface so a number is never unlabelled
better	lower, higher, or None	Tells the results page which direction is stronger. Without it a column would silently sort backwards: context++ and ΔG are better when lower, an ML probability when higher
categories	Required when better is None	A categorical tool must publish its own ranking, best first. The seed analyser publishes 8mer > 7mer-m8 > 7mer-A1 > 6mer, so no frontend has to invent an order
gives_position	Does it report a coordinate	TargetScan and miRTarBase report none, which is why they have no lane in the binding map
slow	Should it be excluded by default	RNAhybrid is 99.9% of a transcriptome-wide job's runtime

Three empty-cell states, which are not the same thing

State	Meaning	How it is determined
not run	The tool was not selected for this job	No row in tool_runs for that tool
not covered	The tool ran but cannot assess this gene	covers() returned false — e.g. TargetScan has no entry for the gene
no site	The tool ran, covers the gene, and found nothing	covers() true, no record produced

A blank cell must never be read as "no target". In the interface these render as "not run", "N.A." and "—" respectively, and the distinction is explained in the report's legend.

7. The four analysis tools in detail

7.1 Seed analyser — written for this project

Source: src/tools/seed.py and src/tools/seed_tool.py. Version 1.0. Kind: prediction. Categorical, so it publishes categories rather than a numeric score.

Algorithm

- Normalise the miRNA: uppercase, strip whitespace, convert RNA (U) to DNA (T).
- Extract the seed: nucleotides 2–8 from the 5′ end (7 nt).
- Build the 6mer target site: the reverse complement of miRNA positions 2–7.
- Find every occurrence of that 6mer in the transcript, including overlapping ones.
- Grade each hit by two extra conditions: m8 (the base before the 6mer complements miRNA position 8) and A1 (the base after the 6mer is an adenine).

Site type	Conditions	Confidence
8mer	m8 match and A1 present	Highest
7mer-m8	m8 match only	High
7mer-A1	A1 only	Moderate
6mer	Neither	Flagged low_confidence, hidden by default

Strictly Watson–Crick. No G:U wobble pairs are permitted. This is what makes it a different measurement from RNAhybrid, which does allow them — see section 7.2.

Input validation rejects anything that is not DNA or RNA, strips a FASTA header if present, and raises rather than proceeding on an empty or malformed sequence. A miRNA shorter than 8 nt raises explicitly instead of producing a truncated seed.

7.2 RNAhybrid 2.1.2-8 — thermodynamic duplex folding

Run live as an external process for every job. Kind: prediction. Score: minimum free energy in kcal/mol, better when lower. Marked slow, so it is excluded from the default tool set.

This tool required a dedicated investigation before the orchestrator could be designed around it. Seven behaviours were found that would each have corrupted results silently. They are recorded in manifests/rnahybrid_spike.md and are summarised here because several are the kind of thing a supervisor will reasonably ask about.

Finding	Consequence if ignored	Rule adopted
FASTA must be line-wrapped	A sequence on one long line loses a nucleotide: 1,918 nt reported as 1,917, 2,000 as 1,998. Every derived coordinate shifts	Always write FASTA wrapped at 60 characters, and assert the reported length equals the length supplied
-m must be passed	Targets beyond roughly 2,000 nt abort with "target too long" and produce no output at all. DICER1 is 10,384 nt	Pass -m at least the length of the longest target
-f 2,7 is required	Without it RNAhybrid ranks by folding energy alone and GC-rich non-seed decoys outrank the real site. On PDCD4 the true 8mer ranked fifth behind four decoys reaching −23.1 kcal/mol	Always pass -f 2,7. RNAhybrid is a scorer for seed-anchored candidates, not a finder
P-values are length-dependent	The same duplex gives p = 1.000 in a 1,918 nt target and p = 0.227 in a 30 nt fragment	Store MFE, never the p-value, and never present p as confidence
No hit is reported as mfe 0.0 at position 0	A naive parser records absence as a perfect-energy hit	Treat mfe 0.0 with position 0 as no hit
Errors go to stderr	"target too long" is invisible to a stdout-only reader	Capture stderr as well as stdout
Positions are 1-based on the sequence supplied	Off-by-one on every coordinate	Transcript coordinate = region start + reported position − 1

G:U wobble pairs, and why they matter

The -f 2,7 flag forces a helix over query positions 2–7 but permits G:U wobble pairs inside it. RNAhybrid therefore reports seed-anchored sites that an exact Watson–Crick matcher rejects. This is a feature — non-canonical sites are part of why RNAhybrid is included — but it changes what "no seed site" means. A gene may carry an RNAhybrid hit and no seed hit, and that combination is informative rather than contradictory.

The worked example, for hsa-miR-21-5p:

```
PDCD4  3'UTR 1790-1813  -18.5  target AUAAGCUA / miRNA UAUUCGAU
                               all Watson-Crick, the canonical 8mer

GAPDH  CDS    724-749   -20.6  target UGAGCUG  / miRNA AUUCGAU
                               two G:U wobbles in the seed helix
```

The wobble-paired site in a housekeeping gene scores stronger than the validated canonical site in the textbook miR-21 target. Folding energy alone would rank GAPDH's coding region above PDCD4. This single observation is the concrete justification for per-tool columns and no unified score.

Choosing the energy threshold from the data

The low_confidence cutoff was not chosen by convention. Genes reached at each cutoff for hsa-miR-21-5p:

Cutoff (kcal/mol)	Genes reached	Share of transcriptome
−15	17,643	91%
−20	11,741	61%
−25	1,696	8.8%
−30	65	0.3%

At −15 RNAhybrid reports a site in 91% of human genes, which is not a prediction. low_confidence is therefore set at −25, where the gene count (1,696) is comparable to TargetScan (2,598) and the seed analyser (6,673). Weaker sites are still stored and still shown when asked for.

Stated caveat: a fixed energy threshold favours long transcripts, which have more opportunities to contain a low-energy duplex. This is the same composition confound measured for seed matches (section 18), and it has not yet been quantified for MFE.

7.3 TargetScan v8.0 — precomputed lookup

Kind: prediction. Score: context++, better when lower. gives_position is false.

At query time the tool looks up the miRNA's accession in targetscan_mirnas to obtain its seed family, then selects every prediction for that seed. It returns, per gene: context++ score, conserved and non-conserved site counts, aggregate PCT, whether the gene is in TargetScan's default set, whether TargetScan's representative transcript is the MANE one, and whether the seed still matches the current miRBase release.

If the seed has changed since v22 the tool emits a warning naming both seeds, which is surfaced in the job record rather than discarded. If the miRNA is absent from TargetScan entirely, it returns no records and a warning saying so — distinct from returning no records because there are no predictions.

A gene with zero conserved sites is flagged low_confidence, and aggregate PCT of exactly 0.000 is stored as null rather than zero, because 0.000 in TargetScan's files means "no conservation evidence" and not "probability zero".

7.4 miRTarBase 10.0 — experimental evidence

Kind: evidence. Score: number of strong functional experiments, better when higher. covers() returns true for every gene, because the absence of a record is itself a meaningful answer — the pair is novel.

Each record carries a status derived from the evidence counts:

Status	Condition	Rendered as
validated_strong	At least one strong functional experiment	Validated (strong evidence)
reported_weak	Weak functional evidence only	Reported (CLIP)
evidence_of_no_effect	Only non-functional records	Reported: no effect

Up to 20 PubMed identifiers are carried with each record so a claim can be followed to its source. A score of 0.0 means "no strong evidence, weak evidence exists" and must never render as a numeric zero — the information is in extra.status, and the interface reads it from there.

8. The orchestrator and the job model

A job is one miRNA

One job is one miRNA against one target set. Several miRNAs become several jobs sharing a batch_id. The consequences are practical: a failure costs one miRNA rather than a whole submission, retries are cheap, and progress is reportable per miRNA.

Execution

- create_job resolves the miRNA and every gene identifier, rejects unknown tools, and writes a queued row. It resolves at the gate so the API and the command line validate identically.
- run_job clears any previous results for that job (so re-running is idempotent), then runs each selected tool in turn.
- A tool that raises is recorded in tool_runs with status 'failed' and the exception text; the other tools still run and their results are stored.
- run_job refuses to report 'ok' on an empty target set.

The background worker

scripts/worker.py drains three queues in dependency order: jobs first, then enrichment runs, then reports. The order is not arbitrary — an enrichment reads a job's results and a report reads both, so finishing what a thing depends on first is never wrong.

next_queued() claims work atomically with a conditional UPDATE, so several workers would be safe. requeue_stale() returns anything left in 'running' by a crashed worker back to the queue on startup.

Why everything slow is queued

A transcriptome-wide RNAhybrid job takes 13 minutes 24 seconds. An enrichment takes about two minutes, most of it waiting on the KEGG server. No HTTP request may hold a connection for that long, so job submission, enrichment and report generation all return 202 Accepted with an identifier to poll. There is one code path for every job, rather than a fast path that behaves differently.

9. The HTTP API

FastAPI over the orchestrator. One SQLite connection per request, opened through a dependency and closed when the request ends. Interactive documentation at /docs.

Method and path	Purpose
GET /health	Liveness only
GET /about	Name, subtitle, description, and every data source with its version
GET /tools	Registry META, so a client never invents a ranking
GET /mirnas/{query}	Resolve a miRNA: accession, mature sequence, seed, TargetScan family
GET /genes/{gene}	MANE transcript length and region bounds
POST /jobs	Resolve, validate, queue → 202
GET /jobs/{id}	Status, timings, per-tool runs
GET /jobs/{id}/results	Filtered, sorted, paginated rows
GET /jobs/{id}/results.tsv	Every row, flattened, with provenance
GET /jobs/{id}/results.xlsx	The same rows as an Excel workbook
GET /jobs/{id}/genes	One page of genes with records and transcript geometry
GET /jobs/{id}/geneset	Live gene count for the pathway tick boxes
GET /jobs/{id}/network	Nodes and edges for the network view
POST /jobs/{id}/enrichment	Queue GO and KEGG → 202; reuses an identical run
GET /enrichment/{run_id}	One run: provenance, tab counts, a page of terms
GET /enrichment/{run_id}/terms.tsv | .xlsx	Terms with the four hypergeometric integers
POST /jobs/{id}/reports	Queue an HTML report → 202
GET /reports/{id}.html	The rendered report, served inline

Two rules the server enforces rather than trusting the client

Sorting reads direction from the tool. sort=score without a ?tool= parameter returns 422. Scores from different tools are not comparable, so the server refuses to pretend otherwise. sort=score on the seed analyser is also a 422, because it reports a category rather than a number; sort=site_type orders by that tool's own published categories.

Errors are structured, not strings. mirna_not_found (404), mirna_ambiguous (409, with the candidate accessions listed), bad_request (422), job_not_found (404).

Bugs this phase exposed and fixed

- create_job did not resolve gene identifiers. The API's first real request scanned zero transcripts and every tool reported "ok, 0 results" — indistinguishable from a true negative. Fixed at the gate so every caller benefits.
- run_job could report success on an empty target set. It now refuses.
- Unknown tool names failed minutes later inside the worker. They are now rejected at creation.

- busy_timeout was set only in the worker, not in db.connect(). Under concurrent access the API could fail on a lock the worker held.

Verified by scripts/test_api.py: 21 assertions, run in-process with no server needed, covering every endpoint, every error path, the sort rules and a known-answer check on PDCD4. A transcriptome-wide run through the API returned 13,920 results for the three fast tools — identical to the figures the command line produced through a different code path.

10. The web interface

One file, web/index.html, generated by scripts/build_page.py from an approved design layout and served by FastAPI at the root of the same process. Same origin, so no CORS is required. No React, no npm, no build step. The only third-party JavaScript is the vendored network library.

The page holds no biology. Site ranking comes from /tools, score direction from each tool's META, and version strings from /about. A tool that was not selected renders "not run", never an em dash. This means adding a fifth tool changes no frontend code.

The page is generated rather than hand-written because the approved design must stay the record of what was approved. build_page.py performs asserted string replacements on it — every replacement checks that its anchor occurs exactly once, so a change to the design that breaks an assumption fails loudly at build time instead of producing a subtly wrong page.

Sections: query, tools run, binding map, the result table, pathway analysis, enriched terms, target network, and a "how to read this page" explainer. The binding map draws exactly one transcript — the one belonging to the selected row — because a position only means something against the sequence it was measured on.

11. Pathway analysis (GO and KEGG)

Gene-set over-representation analysis through R 4.5.0 and clusterProfiler 4.18.4, against org.Hs.eg.db 3.22.0. Every design choice below was measured, not chosen by taste.

The finding that determined the database schema

From one 381-gene list, the annotation covers a different part of the background in each ontology:

Ontology	Genes tested (n)	Background (N)
GO: Biological Process	357	17,116
GO: Molecular Function	362	17,807
GO: Cellular Component	372	18,206
KEGG	190	8,516

All four from the same gene list. KEGG annotates about half the genome. This is why the four hypergeometric integers are stored per term and not once per run, why a KEGG p-value and a GO p-value are not comparable quantities, and why the interface never mixes ontologies in one ranking. A term absent from KEGG may mean "not enriched" or "not annotated", and those cannot be told apart from the p-value alone.

Choosing the gene list

Enrichment needs roughly 200–1,500 genes: fewer and nothing reaches significance, more and everything does. Eleven candidate rules were measured on the same transcriptome-wide miR-21 job:

Rule	Genes
Seed: 8mer only	628
Seed: 8mer or 7mer-m8	2,545
Seed: any 7mer or better	3,480
Seed: any site including 6mer	6,673
Seed 7mer+ outside the 3′UTR	1,244
TargetScan: any score	2,598
TargetScan: conserved site (default)	381
TargetScan: context++ ≤ −0.30	320
RNAhybrid: ΔG ≤ −25	1,805
miRTarBase: strong evidence	124

Rule	Genes
8mer and TargetScan conserved	150

Three results, including two negatives

1. An 8mer alone produces no biology

628 genes, all 628 recognised by org.Hs.eg.db, and zero enriched terms. A canonical 8mer anywhere in a transcript is not evidence of function. This is a clean negative result and the downstream counterpart of the site-level argument for showing four columns instead of one.

2. Conservation alone gives coherent, correct biology

381 genes, all recognised, 340 enriched terms. Read from stored run enr_3e67cd65200d:

GO: Biological Process	Genes	Adjusted p
Mesenchyme development	29	1.9×10^{-7}
Stem cell differentiation	25	4.0×10^{-7}
Lung morphogenesis	13	4.0×10^{-7}
Mesenchymal cell differentiation	24	8.6×10^{-7}

KEGG pathway	Genes	Adjusted p
MicroRNAs in cancer	17	2.2×10^{-5}
MAPK signalling	21	3.6×10^{-4}
Pathways regulating stem-cell pluripotency	13	1.3×10^{-3}
Proteoglycans in cancer	15	2.4×10^{-3}
Ras signalling	16	3.0×10^{-3}

The dominant theme is the mesenchymal and stem-cell programme, and the strongest KEGG pathway in a database that knows nothing about this project is MicroRNAs in cancer. The tool reached that from sequence and conservation alone, with no knowledge of the miR-21 literature.

A correction worth disclosing. An earlier draft of the manifest named epithelial-to-mesenchymal transition as the leading term at 1.9×10^{-7}. That was wrong — the p-value belongs to mesenchyme development, and EMT itself sits at 1.8×10^{-4} with 16 genes: present and consistent, but three orders of magnitude weaker than claimed. The error came from reading the wrong row out of a terminal. Every number above is now queried from the database, which is the reason the database exists.

The EMT family in full: mesenchymal cell differentiation (24 genes, 8.6×10^{-7}), cardiac EMT (8, 1.3×10^{-4}), EMT (16, 1.8×10^{-4}), mesenchymal cell proliferation (8, 5.0×10^{-4}), regulation of EMT (10, 3.5×10^{-3}). The signal is real and spread across the family rather than concentrated in the one term with the famous name.

3. Intersecting with the seed scan adds nothing

The intersection of the 8mer list and the TargetScan conserved list (150 genes) gave 101 terms, best adjusted p 5.4×10^{-4}. Ten random 150-gene samples drawn from the same 381 conserved genes gave 9, 37, 55, 69, 73, 80, 96, 131, 152 and 301 terms. The real list ranks 8th of 11. Empirical $p \approx 0.36$.

The seed filter is not selecting biology; it is only making the list smaller. On reflection this is expected and the earlier reasoning was wrong: TargetScan's predictions are themselves seed-based, so intersecting them with our seed scan is not combining independent evidence. It mostly selects the TargetScan genes whose site happens to be an 8mer.

A caution that came out of the control

Random 150-gene subsets of the same 381 genes produced between 9 and 301 enriched terms — a 33-fold range, with top terms ranging from "ear development" to "MicroRNAs in cancer". Enrichment on a few hundred genes is unstable. Two consequences were adopted: prefer the largest defensible list over the most stringent one, and never read a single enrichment run as a settled result.

Implementation choices in R

- The background is passed explicitly. Omitting it makes clusterProfiler substitute its whole annotation database, which is not the set that was scanned.
- Benjamini–Hochberg correction, terms kept at adjusted $p < 0.05$.
- All three GO ontologies plus KEGG returned in one table with a source column.
- The number of gene ids the annotation recognises is printed, so "no enrichment" can never be confused with "nothing mapped".
- KEGG contacts a live server and is allowed to fail without taking GO down with it.
- A provenance sidecar records R, clusterProfiler and org.Hs.eg.db versions, whether KEGG answered, and when — because the same gene list next year can give different pathways, and this is what will explain it.

Validation of the enrichment storage

scripts/validate_enrichment.py checks three separate claims. Faithfulness: the database matches the TSV that R wrote, exactly, for term ids, descriptions, k and n values, p-values,

adjusted p-values and gene members. Self-sufficiency: every p-value recomputed from the four stored integers, worst discrepancy 7.2×10^{-11}. Reproducibility: compared against an earlier independent run — strict for GO, reported rather than failed for KEGG because KEGG is fetched live. Both stored runs pass, with gene symbols traced 249/249 and 80/80.

12. The target network

A second reading of rows already in the database, not a new analysis. It uses the same ranking function as the results table and the same per-gene summariser as the report and the Excel export, so the three cannot disagree about which genes are shown or which site leads a gene.

Two kinds of edge, meaning different things

Edge	Meaning	Width encodes
miRNA → gene (solid)	At least one tool reported this target	How many of the four tools agree — the only honest cross-tool number available
gene — gene (dotted)	The two genes appear together in enriched terms	How many terms they share

Node colour distinguishes validated (miRTarBase evidence exists) from predicted only. Colour is not a scale: evidence and prediction are different kinds of claim, so they get different colours rather than lighter and darker shades of one.

The combinatorial trap, and the cap that avoids it

A term containing 20 of your genes generates 190 gene-pair edges on its own, linking everything to everything and saying nothing about which genes belong together. Terms holding more than 12 of the displayed genes are therefore skipped for the gene–gene layer, and the interface reports how many were skipped and why rather than hiding it.

What the network made visible

On the nine-gene validation job, two findings are apparent at a glance that take minutes to locate in a table. GAPDH — the negative control — is the smallest node with the thinnest edge and no colour for evidence, because only one tool found it. TPM1 has three tools rather than four, TargetScan being the one missing, for the transcript-choice reason explained in section 14.

A third observation came from the network itself. PTEN, the most heavily validated target in the set with 91 strong functional experiments, carries no gene–gene links at all. The reason is that the enrichment was run on genes with a conserved TargetScan site, and PTEN's site is not conserved, so it was never in the tested list. The pathway result is therefore the story of five genes rather than of miR-21. That is not an error — the filter is defensible — but it is exactly the kind of consequence that is invisible in a table of 232 terms and obvious the moment the data is drawn.

13. The export layer

Three formats for two different readers, all generated from one provenance function, so no two files can claim different versions.

Format	Reader	Design
TSV	R and pandas	Provenance as '#' comment lines, which both skip with one argument. One row per stored record. No JSON in any cell — the stored extra object is spread into named columns. Tab-separated rather than comma-separated because gene descriptions and GO term names contain commas
Excel (.xlsx)	People	Four sheets: results one row per gene with tools across the columns; all sites one row each without sequence strings; provenance as key–value rows; tools run. Header frozen, filter on every column
HTML report	Reading and printing	Queued and stored, never re-rendered. Contains a scope statement, a legend, the gene table, binding maps, the enrichment tables, a network figure and the provenance block. Served inline so Ctrl+P produces a PDF without any PDF toolchain

Faithfulness of a filtered export

The page's region checkboxes, include-6mer toggle and gene search filter in the browser. An export that ignored them would contain rows the screen was hiding, with nothing to indicate it. The export endpoints therefore accept all three, and the provenance block records each one.

Two subtleties in the SQL. A row with no region of its own — miRTarBase evidence is about a gene, not a place inside it — is never hidden by a region filter. And the 6mer filter is scoped to the seed analyser so it cannot silently remove another tool's rows.

Figures in the report are static SVG

The report's value is being one self-contained file that can be emailed. Loading the network library from the server would make it work only while that server runs. Both figures are therefore drawn server-side as SVG — a list of drawing instructions inside the document, which prints at print resolution and needs no JavaScript.

The network layout is deterministic. Starting positions come from node order rather than a random seed, so the same job always produces the same figure — which the interactive physics-based version cannot promise, and which matters for a figure that may be cited.

14. Validation against the published literature

Nine genes, four tools, one job, run through the web interface so the test also exercises the API and the frontend. Eight are miR-21 targets with published functional evidence; GAPDH is a negative control. The job produced 53 records: seed 13, RNAhybrid 26, TargetScan 6, miRTarBase 8.

Gene	Seed	Region	ΔG	context++	PCT	Strong exp.	No effect
TIMP3	8mer ×2	3′UTR	−22.8	−0.675	0.87	15	0
PDCD4	8mer	3′UTR	−18.5	−0.503	0.85	59	6
SPRY2	8mer	3′UTR	−19.4	−0.385	0.69	6	1
RECK	8mer	3′UTR	−24.9	−0.378	0.63	15	2
BCL2	7mer-m8	CDS	−19.4	−0.117	0.27	9	3
PTEN	7mer-A1	3′UTR	−20.2	−0.012	none	91	23
TPM1	7mer-A1	3′UTR	−22.1	none	—	6	1
MARCKS	none	—	−21.1	none	—	2	2
GAPDH	none	—	−20.6	not covered	—	not reported	—

Recall on the eight published targets

Tool	Recovered	Note
miRTarBase	8 / 8	And GAPDH correctly absent
Seed analyser (ours)	7 / 8	Only MARCKS has no canonical site
TargetScan	6 / 8	Misses TPM1 and MARCKS
RNAhybrid	9 / 9	Including the negative control — sensitive, not selective

miRTarBase recovering 8/8 is a check on the loader and the name reconciliation — 91,657 aliases across historical miRBase releases — not on the prediction. miRTarBase is the reference here.

Finding 1: prediction strength does not track experimental support

```
by context++     TIMP3 > PDCD4 > SPRY2 > RECK > BCL2 > PTEN
by experiments   PTEN  > PDCD4 > RECK  > TIMP3 > BCL2 > SPRY2
```

PTEN is last by prediction and first by evidence: 91 strong experiments across 20 papers, context++ −0.012, and no conserved site. A single combined score would have averaged the most-validated target in the set down to nothing. This is the concrete case for per-tool columns.

Caveat, stated plainly: experiment counts measure how much a pair has been studied, and PTEN is a famous gene. The disagreement is real; the reason for it is not only biology.

Finding 2: TargetScan misses TPM1 because of transcript choice, not biology

TPM1 is present in targetscan_genes but has zero rows for the miR-21 seed. All TargetScan predictions are loaded, not only conserved ones, so any 7mer-A1 in the 3′UTR TargetScan scored would have produced a row. Our scanner finds a 7mer-A1 at 1,126–1,132 on the MANE Select transcript NM_001018005.2, and the pair is published (Zhu et al., 2007).

The site therefore lies outside the 3′UTR TargetScan chose. TPM1 is heavily spliced. This is the 38.7% transcript mismatch measured during the TargetScan load, landing on a validated target — and a direct argument for anchoring on MANE Select.

Finding 3: BCL2's strongest canonical site is in the CDS

A 7mer-m8 at 944–950, in coding sequence. TargetScan reads 3′UTRs only, so it scores BCL2 on a different, weaker site (−0.117, PCT 0.27) while the pair carries nine strong experiments. A 3′UTR-only tool cannot see this class of site, which is the argument for the planned ML model covering all three regions.

What this validation is not

- Nine hand-picked positives are not a benchmark. There is no random negative set, so no precision, no false-positive rate and no ROC curve.
- There is no independent replication of miRTarBase; it is the evidence source.
- A real benchmark needs a defined miRNA set, all miRTarBase strong pairs as positives, matched random genes as negatives, and recall and precision per tool. It is planned as its own phase and its scripts are written.

15. Measured performance

Operation	Time
Three fast tools, transcriptome-wide (19,363 transcripts)	~1 second
Adding RNAhybrid, parallel across 12 workers	~4 minutes
RNAhybrid serial, same job	13 min 24 s
GO + KEGG enrichment, 381 genes	~125 seconds
HTML report, 9 genes	~0.01 seconds
Loading 72 Mnt of transcript sequence from SQLite	0.06 seconds
Full transcriptome seed scan, one miRNA	0.17 seconds

Transcriptome-wide results for hsa-miR-21-5p: 58,954 records — seed 10,254 rows across 6,673 genes; RNAhybrid 45,034 rows across 17,671 genes; TargetScan 2,598; miRTarBase 1,068.

The parallelisation bug, and why it is worth recording

RNAhybrid was parallelised from 803 seconds to 231 seconds — 3.5× on 11 cores — with output verified identical record for record (SHA match on 45,034 rows).

The first parallel version was slower than serial. It collected worker output through pipes. A pipe holds roughly 64 kB; once full, the writing process blocks until someone reads it. Eleven workers therefore sat frozen while the first was drained. Each worker now writes to its own file.

The instructive part is why it was not caught earlier: a 2,000-transcript test scaled almost 4× and looked correct, precisely because its output stayed under the buffer size. The bug was only reachable at full scale. This is now a standing argument for testing performance work at production scale rather than on a convenient subset.

16. Problems faced at each stage, and how each was solved

This section collects, in one place, every problem that changed a design decision. Each entry names the symptom, the diagnosis and the fix.

Data layer

Problem	Diagnosis	Solution
243 human miRNA seeds differ between miRBase 22 and 23	5′ end re-annotation renumbers positions 2–8; not different miRNAs	Key miRNAs by accession, join TargetScan by seed family, flag changed seeds in the output
TargetScan's miRBase release was unknown	Measured seed agreement against both R22 and R23: 99.84% with R22	Treat TargetScan as a v22-era resource and bridge by accession, reporting the 2.8% that cannot be bridged
TargetScan and MANE use different gene identifiers	Measured three join keys; Ensembl gene best at 18,187	Resolver cascade (Ensembl gene → symbol → alias) reaching 95.5%
TargetScan scores a different transcript for 38.7% of genes	Its representative transcript is chosen by 3P-seq, not by MANE	Join and display at gene level only; never show a TargetScan position
22.5% of miRTarBase evidence rows unresolvable	Papers from 2010–2013 use pre-suffix names; miRBase serves only v20–v22	Export historical names from miRBaseConverter across 22 releases; residual falls to 4.6%
149,049 rows still unresolved after that	They are miRNAs deleted in Release 23	Report as retired, not as absent — a fact about the namespace, not about the data
miRTarBase page says Release 11.0 but serves 10.0	Every link resolves to /10.0/; /11.0/ returns 404	Record 10.0, which is what is loaded

RNAhybrid

Problem	Diagnosis	Solution
Sequence lengths came back one or two nucleotides short	Unwrapped FASTA on a single long line	Wrap at 60 characters and assert the reported length matches
Long transcripts produced no output at all	"target too long" written to stderr, invisible to a stdout reader	Pass -m, and capture stderr
Decoy sites outranked the real 8mer	Ranking by folding energy alone; the true site was fifth	Pass -f 2,7 to force a seed-anchored helix
Absence of a hit looked like a perfect hit	No hit is reported as mfe 0.0 at position 0	Treat that combination as no hit
Parallel version slower than serial	64 kB pipe buffer blocked eleven workers	Each worker writes its own file
A housekeeping gene	G:U wobbles permitted inside	Keep it — but never combine scores across

Problem	Diagnosis	Solution
outscored the validated target	the forced helix	tools; this is the evidence for that rule

Orchestration and API

Problem	Diagnosis	Solution
Every tool reported "ok, 0 results" on the API's first request	create_job never resolved gene identifiers; the CLI had been doing it by hand	Resolve at the gate, so API and CLI validate identically
A job could report success having scanned nothing	No check on an empty target set	run_job refuses to report ok when the target set is empty
Unknown tool names failed minutes later in the worker	Validation happened at run time	Reject at creation
Occasional database lock errors from the API	busy_timeout set only in the worker	Set it in db.connect(), which every caller uses

Export and reporting

Problem	Diagnosis	Solution
Report HTML unreachable at its URL	FastAPI matches routes in declaration order; /reports/{id} swallowed "rep_abc.html"	Declare the more specific .html route first
A 17-page report from a nine-gene job	Every one of 232 GO:BP terms printed	Cap at 20 terms per ontology, state the full count, point to the TSV
Excel sheet unreadable: 33 columns of alignment strings	Long-format rows designed for R opened in a spreadsheet	Pivot to one row per gene with tools across the columns; move per-site detail to a second sheet
JSON visible inside TSV cells	The stored extra object was written verbatim	Spread it into named columns; no export contains JSON
Worker crashed if started before the API	It opened a raw connection without applying the schema	Connect through db.connect(), which applies the schema
Streaming export would fail mid-download	Since FastAPI 0.106 a yield dependency closes before the streaming body is sent	Materialise the rows before streaming; noted for a future true-streaming export

17. Design decisions and the reasoning behind them

Decision	Reasoning
No unified score, ever	Tool scores use different units and different scales. The PTEN case (section 14) shows a combined score would have ranked the most-validated target in the set last. Users filter and compare per tool instead
Key miRNAs by accession, not name	243 human seeds changed between two consecutive miRBase releases; names are not stable identifiers
MANE Select as the single transcript backbone	A position means nothing without the sequence it was measured on. One agreed transcript per gene makes two tools' positions comparable
TargetScan joined at gene level	Its representative transcript differs from MANE for 38.7% of genes, so its positions are not comparable with ours
Seed matching computed live, never precomputed	Measured at 0.17 s for a full transcriptome scan. A precomputed miRNA × transcript matrix would be large, quickly stale, and unnecessary
RNAhybrid results cached after computation	803 s serial, 231 s parallel. Slow enough to be worth storing
Everything slow is queued	A 13.5-minute job cannot be an HTTP request. One code path rather than a fast path that behaves differently
Nothing is ever deleted	Old runs produced old scientific results. Staleness is derived by comparison, not by destruction
Store the four hypergeometric integers per term	Each ontology annotates a different part of the genome, so one pair per run would make every p-value unrecomputable
Background is every MANE Select gene, always	The 3′UTR-length alternative was measured: 18,622 versus 19,363 at ≥50 nt, a 3.8% difference — too small to justify the extra explanation
Default gene list is TargetScan conserved sites	Measured against ten alternatives. An 8mer-only list gives zero enriched terms; the intersection with the seed scan adds nothing (empirical $p \approx 0.36$)
6mer sites stored but hidden by default	They are real matches but poor predictors. Hiding is a display decision; discarding would be a data decision
The frontend holds no biology	Ranking comes from /tools, direction from META, versions from /about. Adding a tool changes no frontend code
Vendor the network library	Works offline, survives a CDN outage, and a pinned local copy is the only version claim that stays true
Report figures are static SVG	Keeps the report one self-contained file, prints at print resolution, and gives a deterministic layout for a citable figure

18. Known limitations, stated plainly

Scientific

- No benchmark has been run. Validation shows recall on known positives only; precision and false-positive rate are unmeasured. The scripts exist.

- Region counts must not be read as binding preference. Measured: 6mer density is equal across regions (real 3′UTR:CDS = 1.04×, shuffled = 0.94×), and for hsa-miR-21-5p the real 7mer+ enrichment of 2.44× sits inside the range produced by shuffling the same six bases (median 1.62×, range 0.72–2.80). Counting sequence matches cannot demonstrate regional preference; functional 3′UTR enrichment is established from repression data, not match counts.
- The 3′UTR-length bias in the enrichment background is documented, not fixed. miRNA targets have long 3′UTRs, and long-3′UTR genes are enriched for transcription factors and developmental regulators. Fixing this needs length-matched sampling.
- The RNAhybrid energy threshold favours long transcripts, and that confound has not been quantified for MFE.
- Enrichment on a few hundred genes is unstable — random subsets of the same list gave a 33-fold range in term count.
- Experiment counts measure attention as well as biology. A famous gene accumulates papers.

Data coverage

- 184,409 miRTarBase evidence rows remain unresolved; 149,049 of them are miRNAs deleted in Release 23 and are correctly reported as retired.
- 287 human miRNAs postdate TargetScan v8.0 and therefore have no TargetScan prediction.
- 73 TargetScan accessions (2.8%) cannot be bridged to Release 23 at all.
- 881 TargetScan genes (4.5%) did not resolve to a MANE gene.

Engineering

- No authentication, no rate limiting and no CORS configuration. Adequate for local use; not for a public server.
- The API declares no response models, so the interactive documentation shows a generic string for every 200 response body.
- A missing RNAhybrid binary surfaces as a raw FileNotFoundError rather than a readable message.
- There is no single build_all script to rebuild the database from raw downloads in one command.
- The per-tool label field is still deferred; the frontend parses tool-specific JSON in a few places instead.

Portability

Users need only a web browser, on any operating system. The server is effectively Linux-only because of the RNAhybrid binary, which has no official Windows build; everything else in the stack is cross-platform. Docker is the intended answer and belongs to the deployment phase — it would also make the version claims in the provenance block reproducible rather than merely recorded.

19. What is not built yet

Component	State	What it needs
Benchmark	Designed; scripts written, not run	Positives from miRTarBase strong-evidence functional pairs restricted to genes with a MANE transcript; negatives both 3′UTR-length-matched and plain random, reported side by side; precision, recall and false-positive rate per tool. miRTarBase itself is excluded from being benchmarked, since it would be scoring its own answer key
Machine-learning model	Blocked pending supervisory decision	Agreement on the feature set and the training dataset. The settled design computes its own local duplex energy and accessibility in a short window around each candidate site, so it stays independent of RNAhybrid
Differential expression (DESeq2)	Scoped as optional, not started	File upload for a count matrix and design, new schema, a new R execution path
Deployment	Not started	Docker image containing Python, R, Bioconductor and the RNAhybrid binary; a host with a real URL; authentication
miRanda	Dropped	Would need a hand-built binary for a second opinion of the same kind RNAhybrid already gives

A stated limitation of the planned benchmark, worth raising before it is run: there are no true negatives. "False positive" will mean "predicted but not reported", which is partly a statement about how much attention a gene has received rather than about biology.

20. How to reproduce every result in this document

Running the system

```
cd ~/mirna-tool && source venv/bin/activate

# terminal 1 — the API and the web interface
uvicorn api:app --app-dir src --port 8000

# terminal 2 — the background worker
python scripts/worker.py

# then open http://localhost:8000/
```

The validation job

```
python scripts/run_job.py hsa-miR-21-5p PDCD4 PTEN RECK TPM1 SPRY2 \
    TIMP3 BCL2 MARCKS GAPDH \
    --tools seed,targetscan,mirtarbase,rnahybrid
```

Verification scripts

Script	Checks
scripts/test_api.py	21 assertions across every endpoint, error path and sort rule
scripts/validate_enrichment.py	Database against R's own output; p-values recomputed from stored integers; reproducibility against an earlier run
scripts/check_rnahybrid_parity.py	Parallel output identical to serial, record for record
scripts/ check_rnahybrid_positions.py	Every reported position rebuilt from the alignment and compared against the transcript
scripts/composition_control.py	Region enrichment against shuffled-sequence controls
scripts/enrich_control.py	Enrichment of the intersection list against random subsets of the same size
scripts/check_hypergeom.py	p-values recomputed independently of R

Where the evidence lives

Every finding quoted in this document is recorded in manifests/: targetscan_version_finding.md, mirtarbase_finding.md, region_composition_finding.md, rnahybrid_spike.md, phase5_design.md, phase6_api.md, validation_mir21.md and pathway_design.md. Stored enrichment runs can be queried by identifier — enr_3e67cd65200d (381 conserved genes) and enr_0ac52b947705 (the 150-gene intersection) — so any number here can be re-derived rather than taken on trust.

21. Anticipated questions and short answers

Question	Answer
Why not produce a single combined score, like some other tools?	Because the four tools measure different quantities in different units. Concretely: PTEN has 91 strong functional experiments and a context++ of −0.012. Any averaging ranks the best-validated target in the set last. The approach follows miRWalk (own model scored separately, other tools shown alongside) rather than mirDIP (normalised fusion)
Why MANE Select rather than all transcripts?	A position is only meaningful against the sequence it was measured on. One agreed transcript per gene makes the seed analyser's coordinate and RNAhybrid's coordinate directly comparable. The cost is visible in the TPM1 case and is stated
Why is TargetScan shown without positions?	Its representative transcript differs from MANE Select for 38.7% of genes, so a position would refer to a different sequence. Gene-level join, no position — a measured decision, not a simplification
Is the 8mer count evidence of 3′UTR targeting?	No. Measured against shuffled controls, the real seed's enrichment falls inside the range produced by shuffling the same six bases. Region counts reflect composition, not preference
How do you know the enrichment numbers are right?	Three separate checks: the database matches the TSV R wrote exactly; every p-value recomputes from the four stored integers to within 7.2×10^{-11}; and results reproduce against an earlier independent run
What happens when a tool fails?	It is recorded in tool_runs with status 'failed' and the exception text. The other tools still run and their results are stored. A blank cell is never ambiguous — the interface distinguishes not run, not covered and no site
Can the results be reproduced next year?	The stored runs, yes — nothing is deleted, and every version is recorded. A new KEGG query, no: KEGG is fetched live, which is why kegg_fetched_at is stored
What would you do differently?	Test the parallelisation at full scale rather than on a 2,000-transcript subset; the pipe-buffer bug was only reachable at production size. And query results from the database rather than reading them out of a terminal — one enrichment figure was misreported that way and had to be corrected
What is the weakest part of the project?	The absence of a benchmark. Recall on nine hand-picked positives says nothing about precision or false-positive rate. The design is settled and the scripts are written

Prepared from the project's own manifests, database and source. Every figure is re-queryable; every decision named here has a recorded measurement behind it.